Introduction to Histology

DEFINITION

- Histology is the branch of anatomy that deals with the study of tissues in animal and plants.
- It deals with the structure and its function.
- This subject itself intertwines the discipline in cell biology, biochemistry, physiology and pathology.

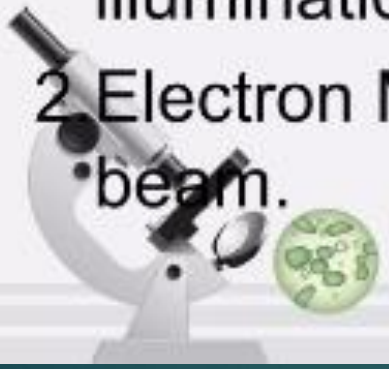
www.bpt.info

MICROSCOPY

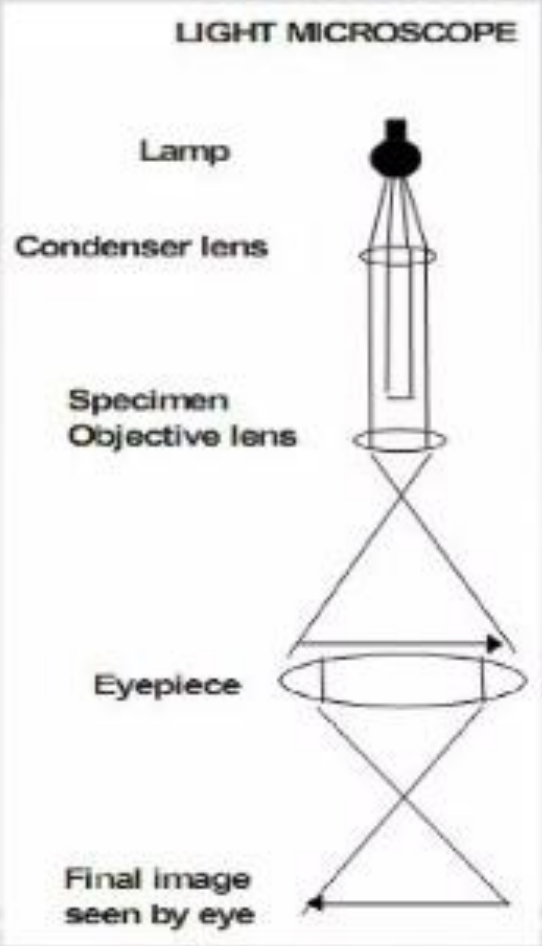
- Is the method used in the study of cells and its structure using microscope.

<u>Major Kind of Microscope</u>

1. Light Microscope – Utilizes light for illumination.
2. Electron Microscope – Uses an electron beam.



LIGHT MICROSCOPE



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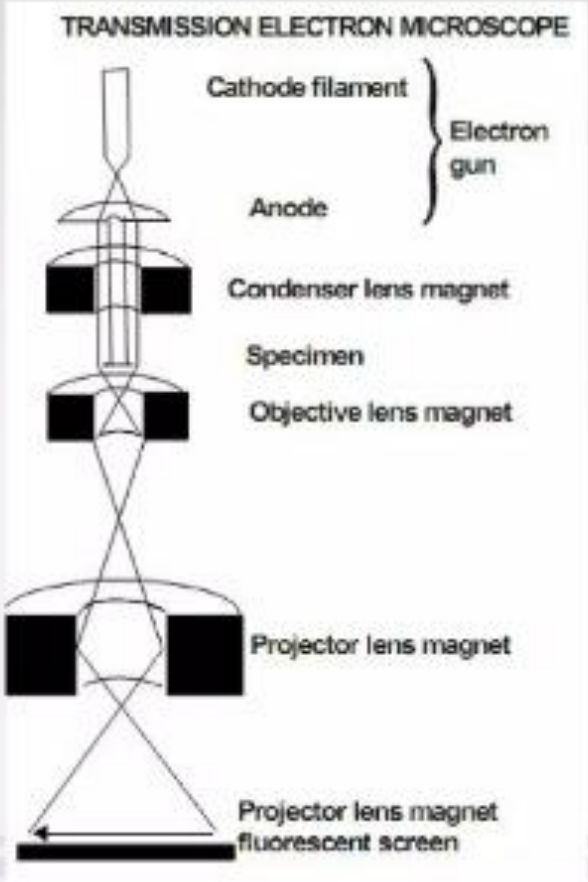
TRANSMISSION ELECTRON MICROSCOPE



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3 IMPORTANT PARAMETERS IN MICROSCOPY

1. Magnification – It is the ratio between the size of an image produce by the microscope and its actual size.
2. Resolution – A measure of clarity of an image.
3. Contrast – The ability to visualized a particular cell structure depending on how different it looks from an adjacent structure.

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TISSUE PREPARATION

1. *Fixation* – treatment of tissue with chemical agent.
2. *Dehydration & Clearing* – removal of water from tissue sample.
3. *Embedding* – Infiltration of tissue sample with paraffin.
4. *Sectioning* – Cutting tissue sample by section into specific equal increments.
5. *Mounting and Staining* – Placing the tissue sample on adhesive glass slides.

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FIXATION

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MOUNTING AND STAINING

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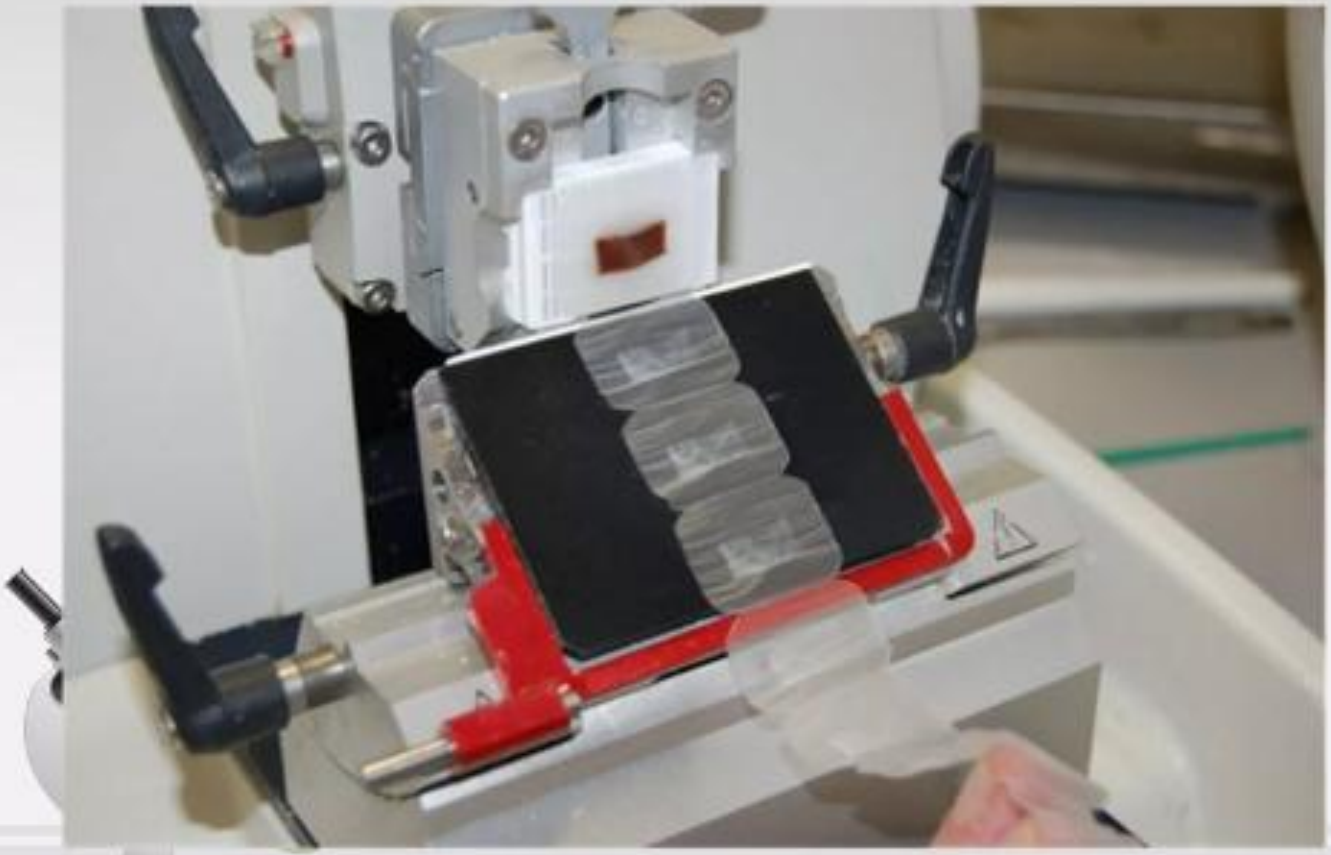
SECTIONING



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Cell division

III. Division of the Cell

1. Before the cell gets too large it divides into two "__daughter__" cells

2. The process is called __cell__ __division__.

Cell division solves the problem of cells getting too large by __Increasing size__ while __reducing volume__ (more small cells instead of one large cell)

3. Before cell division, the cell __replicates__, or copies, its entire DNA.

a. This solves the problem of __information storage__ because each daughter cell gets a complete set of __genetic__ information

4. Each cell has an increase in ratio of surface area to volume that allows for more efficient __exchange of materials__ with the environment.

10-2 Cell Division

Key Concepts: What are the main events of the cell cycle?

What are the four phases of mitosis?

All cell division must involve the replication of DNA before cell division so the genetic information can be transferred to the daughter cells.

In Prokaryotes the rest of cell division is simply to divide the contents.

In Eukaryotes the division is more complex it occurs in two stages:

a. Mitosis: division of the nucleus

b. Cytokinesis : division of the cytoplasm

Unicellular organisms use mitosis and cytokinesis to _reproduce_____.
This is a type of ___asexual_____ reproduction producing two identical daughter cells from _one___ parent cell.

Multicellular organisms use mitosis and cytokinesis to _add new cells to the organism_____ _for growth and development_____

I. Chromosomes

1. Chromosomes are composed of DNA and proteins (histones) and carry the genetic information in eukaryotic cells.

a. Each species of organism has a specific number of chromosomes.

. Humans have 8 chromosomes

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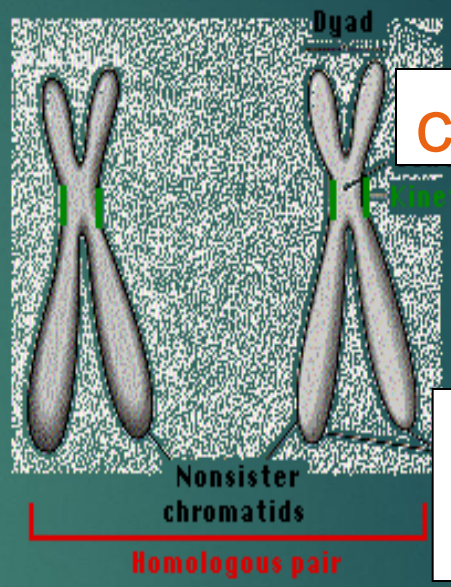
b. Chromosomes are not visible except during cell division. (known as chromatin when not dividing)

c. Replication (copying) of the DNA occurred before cell division therefore every chromosome is actually two identical " sister " chromatids

d. Each pair of chromatids is connected to each other at an area called the centromere, usually located near the center of the chromosome.



centromere

Sister chromatids

http://www.biostudio.com/demo_freeman_dna_coiling.htm

II. The Cell Cycle

1.Cell Cycle: the series of events the cell goes through as they _grow and divide_____ separated by periods of “in-between” time called __interphase__.

2. During the cell cycle the cell __grows___, prepares for ___division__, and divides to form two daughter cells, each of which then _begins the cycle again_____.

3. _Four____ phases of the cell cycle

III. Events of the Cell Cycle

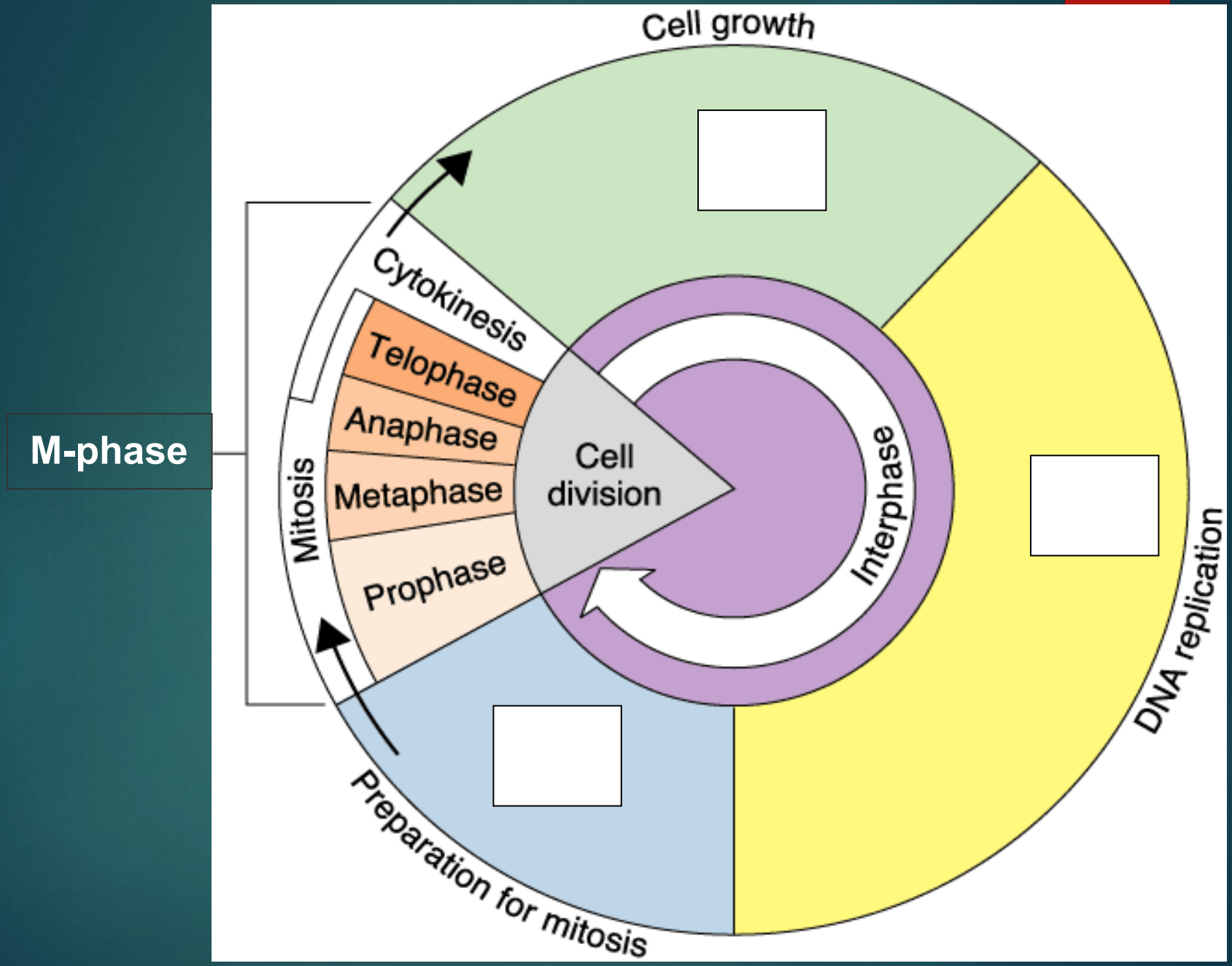
a. **G1 phase**: cell growth, increase in size and synthesize new proteins and organelles.

b. **S phase**: chromosome replication (synthesis)

c. **G2 phase**: shortest of the phases, organelles and molecules required for cell division are produced.

d. **M Phase**: mitosis and cytokinesis

G1, S, and G2 are all taking place during interphase : the phase between divisions.


M-phase
Cell growth
Cytokinesis
Telophase
Anaphase
Metaphase
Prophase
Mitosis
Cell division
Interphase
DNA replication
Preparation for mitosis

III. Mitosis

a. The mitotic phase can be sub-divided into four phases __prophase__, __metaphase__, __anaphase__ and __telophase__ (PMAT).

b. Mitosis is strictly __nuclear__ division.

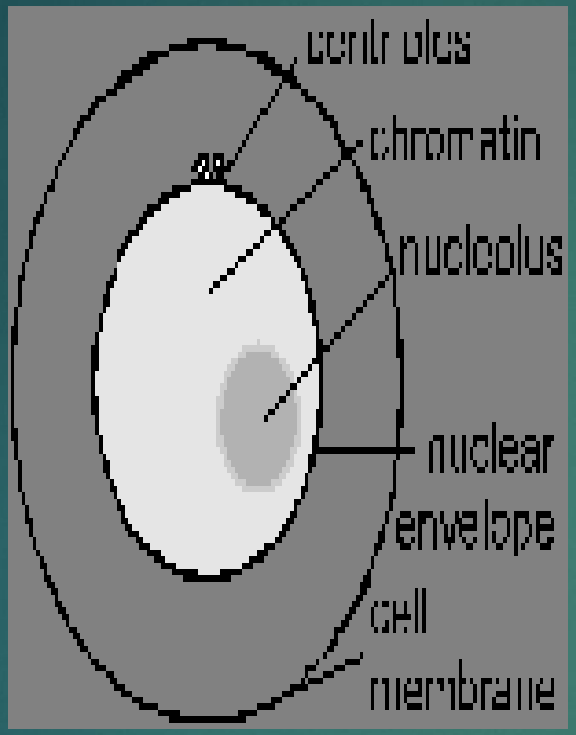
c. Mitosis is followed by cytoplasmic division, or __cytokinesis__, to complete cell division

d. Mitosis results in two “daughter cells”, which are _identical_ to each other, and is used for growth and asexual reproduction.

e. The growth and synthesis phases are collectively called interphase (i.e. in between cell division).

f. The only source of genetic variation in the cells is via ___mutations___.

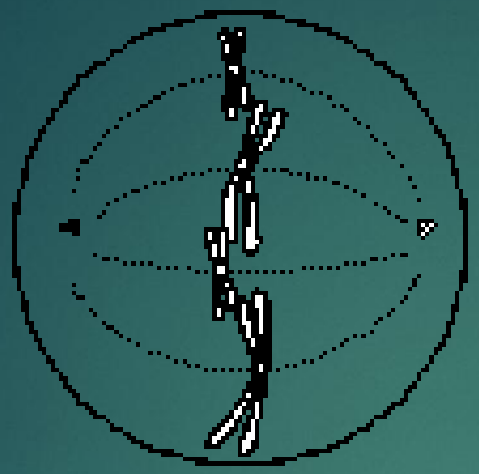
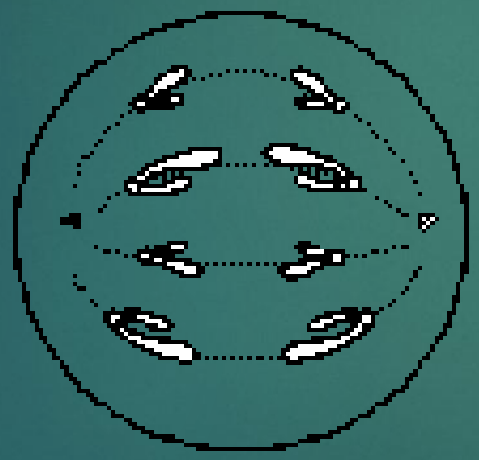
Interphase Stage **between** division

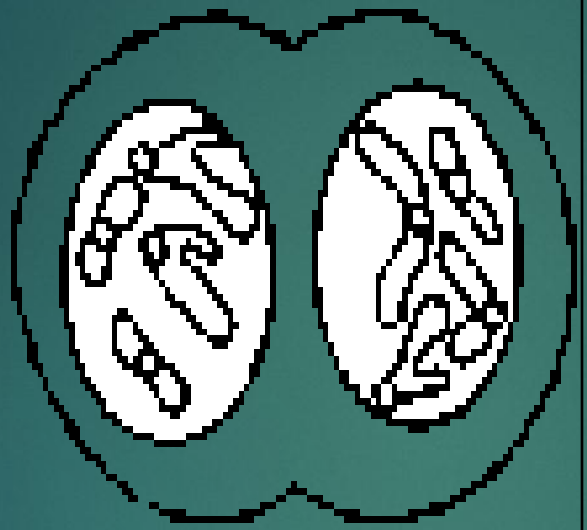


- This is when the cell is not dividing, but is carrying out its normal cellular functions.
- Chromatin is not visible
- DNA, histones and centrioles all replicated
- Replication of cell organelles like mitochondria, occurs in the cytoplasm.
- G1, S and G2 phases are occurring

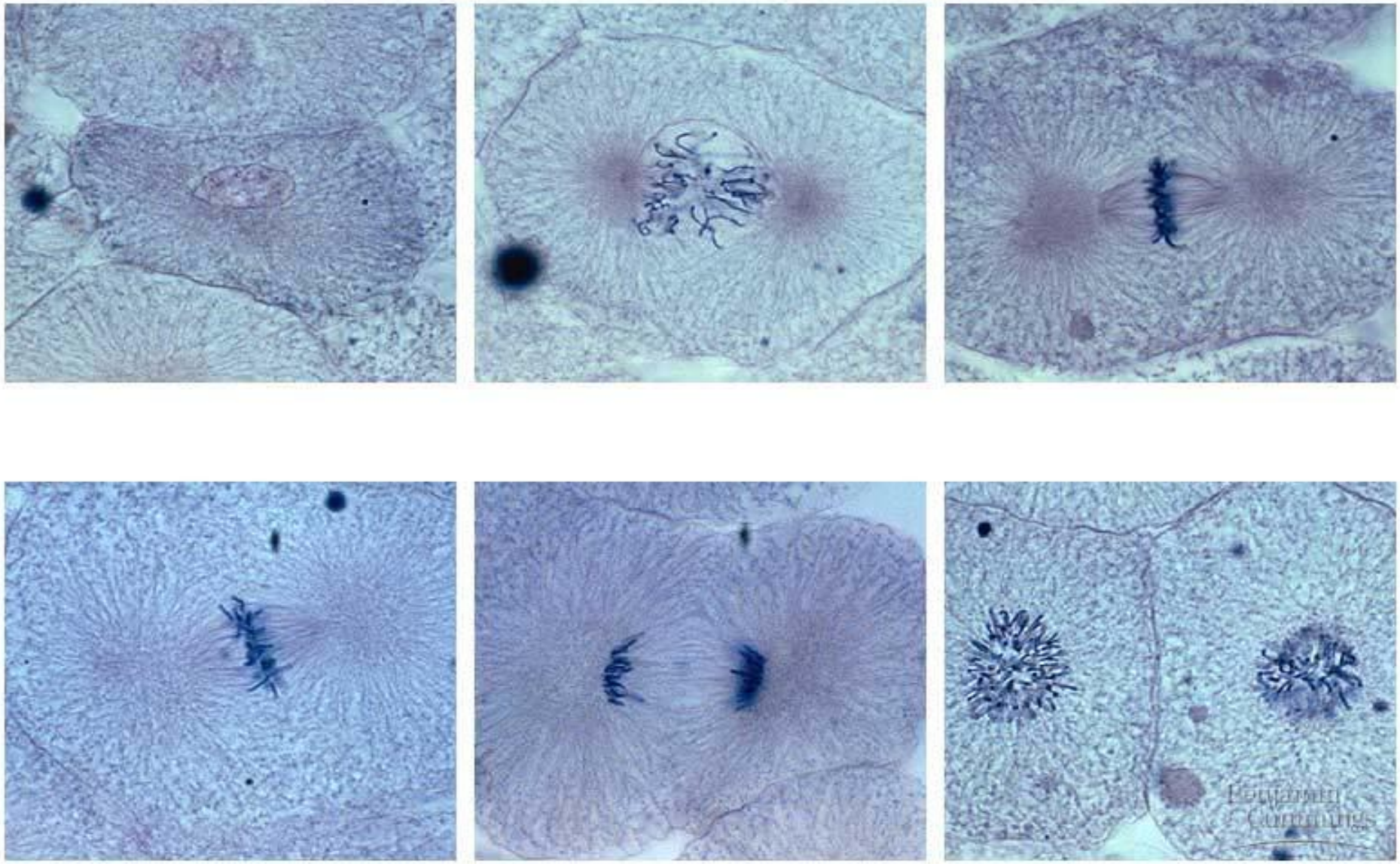
Prophase
Beginning of mitosis (nuclear division)

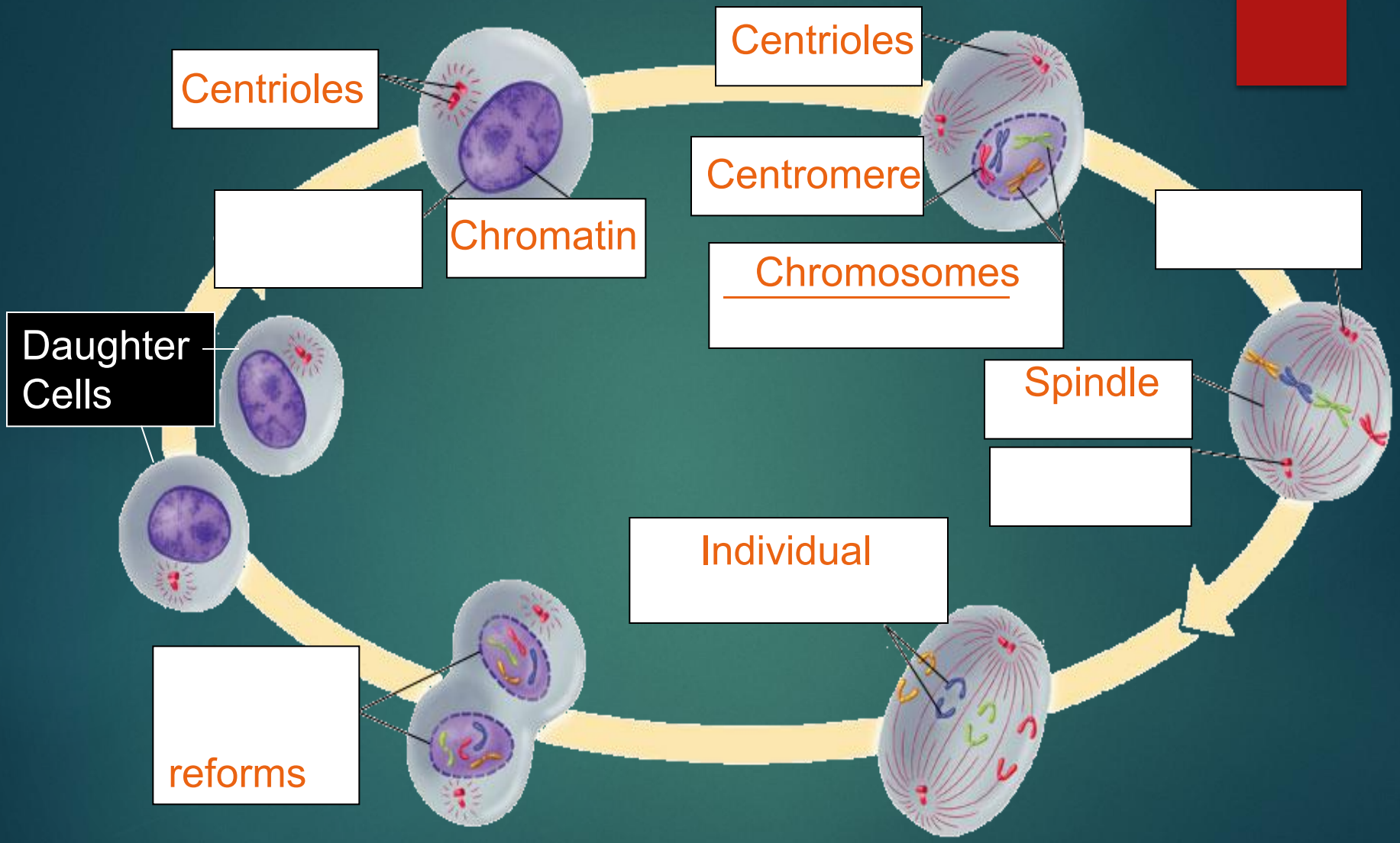
- chromosomes condense and become visible.
- Due to DNA replication during interphase, each chromosome consists of two identical sister chromatids connected at the centromere
- centrioles move to opposite poles of cell (in animal cells)
- Nucleolus disappears
- Spindle fibers begin to form and attach to chromosomes near the centromere
- Phase ends with the breakdown of the nuclear membrane

Metaphase (middle)		•Spindle fibers connect centrioles to chromosomes •Chromosomes align along equator of cell.
Anaphase chromosome separate		•Centromeres split, allowing sister chromatids to separate •Chromatids move towards opposite poles, centromeres first, creating a “V” shape

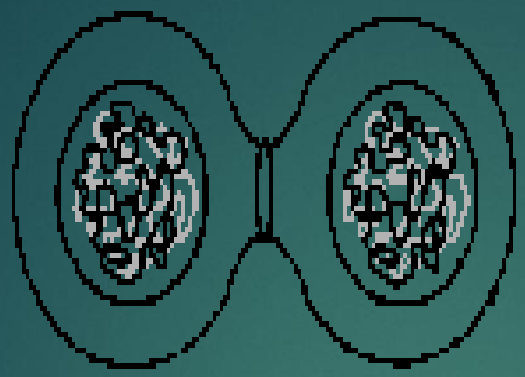
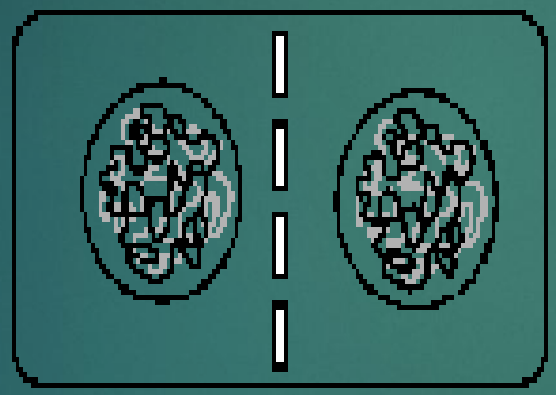
Telophase New nuclei form		•Spindle fibers disperse • Nuclear membranes form around each set of chromatids •Nucleoli reform •End of nuclear division

Animal Cell Division in White Fish Blastula





http://highered.mcgraw-hill.com/sites/0072437316/student_view0/chapter11/animations.html#

Cytokinesis		•In **animal cells** a cleavage furrow forms, which splits the cell in two.
Cytoplasmic division New daughter cells form		•In **plant cells** vesicles move to the equator, line up and fuse to form two membranes called the cell plate. A new cell wall is laid down between the membranes, which fuses with the existing cell wall.

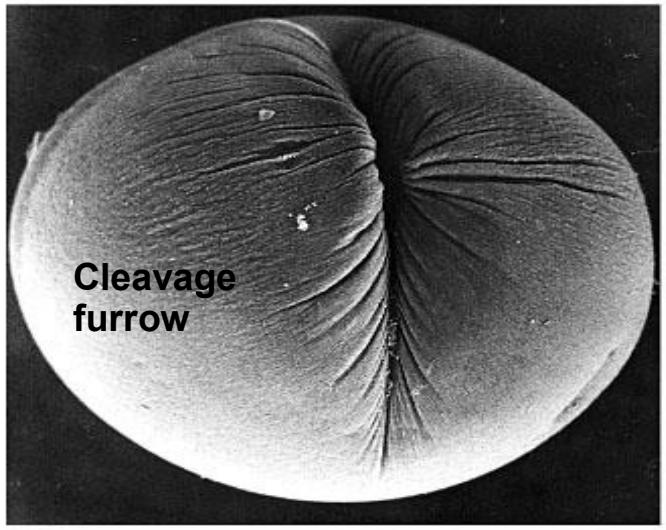
V. Cytokinesis:

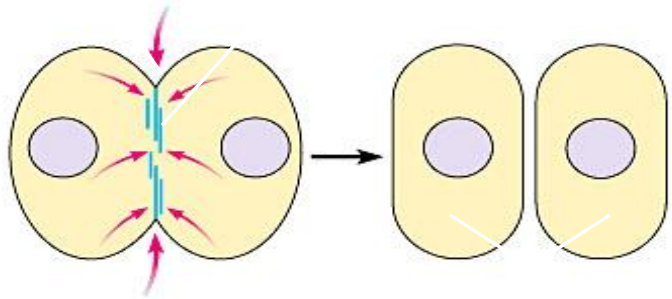
a. Division of the __cytoplasm__

b. End of the __cell__ __cycle__

c. Production of two __identical__ daughter cells

d. Different in __plants__ and __animals__ cells

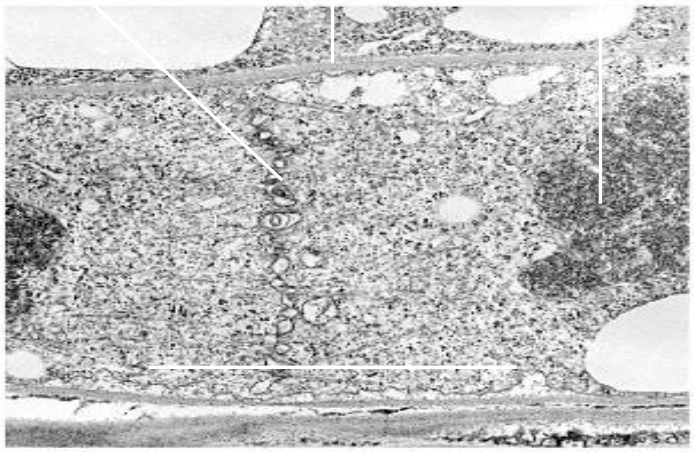


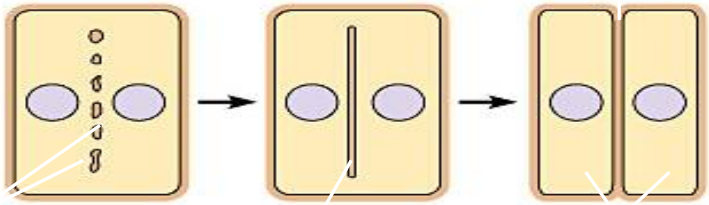


Cell plate
forming

Wall of
parent cell

Daughter
nucleus





Vesicle
cell wall material

cells

Regulating the Cell Cycle

Multicellular organisms control cell growth and division very carefully (way to increase number of cells and size of organism)

This provides the replacement of cells that wear out or are broken down.

II. Cell Cycle Regulators

1. Scientists wondered what ___controlled___ cell division.
2. Tim Hunt and Mark Kirschner discovered that cells in mitosis contained a protein that when injected into a cell would cause the formation of spindle fibers.
3. Protein group known as __cyclins_____ regulates the cell cycle.
4. They rise and fall in time with the _cell cycle_.

III. Uncontrolled Cell Growth

A. Cancer

1. Cell Growth is so controlled because when it is not controlled things go very, very wrong.

2. Cancer is a disorder in which some of the body's own cells lose the ability to control growth.

3. Cancer cells do not respond to the ___signals___ that regulate the growth of most cells.

4. When cells divide ___uncontrollably___ they form masses of cells called ___tumors___ that can damage the surrounding tissue.

5. Cancer cells can break off and ___spread___ throughout the body disrupting normal activities and causing serious medical problems or even death.

B. Causes of Cancer

1. Smoking _tobacco_____
2. _Radiation_____ exposure
3. _Viral Infection_____
4. _Genetic_____ defect

C. Cancer is a disease of the _cell cycle_____.

http://www.gardasil.com/

Breast Cancer Cell